An Introduction to Qubes

Conventional security methods like the use of antivirus software or firewalls are no longer capable of stopping sophisticated malware attacks. Qubes overcomes this flaw by offering secure, compartmentalised virtual machines to run various aspects of the user's digital life. Thus, if a system is compromised by browsing through an untrusted website, banking activities can still be done securely in a VM which is isolated in another compartment.

Qubes OS is a compartmentalised operating system with high security and isolated virtual machines (VMs). It takes the approach of compartmentalisation to deliver security. The OS is open source and freely available at *https://www.qubes-os.org*. Installation is a very easy process, involving just a few clicks.

OS level security

Popular operating systems like Windows or OS X suffer from security issues due to their pre-installed dependencies. For example, opening an innocent portable executable (PE) downloaded from the Internet without knowing about the actual functionality of the particular PE, allows malware to execute on your computer. Once the malware is executed, the entire OS is compromised. Traditional security systems like antivirus software and firewalls are not capable of detecting present day attacks/ intrusions. The Qubes OS provides security services at the operating system level; hence, it is called a security-oriented OS (Figure 1).

Qubes allows users to define their own security domains, which are implemented as lightweight VMs or AppVMs. A person can have personal, work-related, shopping, banking and random AppVMs and can use the applications within those VMs. However, these applications are well isolated from each other. Qubes also supports secure copy-and-paste and file sharing between the AppVMs. The OS executes all the applications in isolated VMs in order to achieve security without being compromised. For example, a user can download any type of file from the Internet and can execute it in one compartment, and the same user can do bank transactions securely in another compartment. Qubes supports inter-VM secure file transfer.

Virtualisation in Qubes OS

Qubes OS adopts an approach called software compartmentalisation, where Type1 virtualisation is achieved by executing the isolated VMs directly on the hardware using the XEN hypervisor (a bare metal hypervisor). This is very different from conventional VMs like Vbox or VMware (hosted hypervisors), and an attacker must be capable of disrupting the hypervisor in order to compromise the entire system, which is immensely difficult.

Dom0

Dom0, an active desktop manager, handles the entire login sessions and is more trusted than other domains. It is the heart of the entire security management. Dom0 is used for running window managers. Due to its overarching importance, connectivity to Dom0 is restricted and the user is never allowed to run an application in this interface.

To open a window in Dom0, go to *Start* → *SystemTools* → *Konsole*

To view various command line tools, please execute the following command:

```
Qvm -ls
```

For managing AppVMs and TemplateVMs, a GUI based 'Qubes Virtual Manager' is used (Figure 2).

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Start → SystemTools → Qubes manager
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